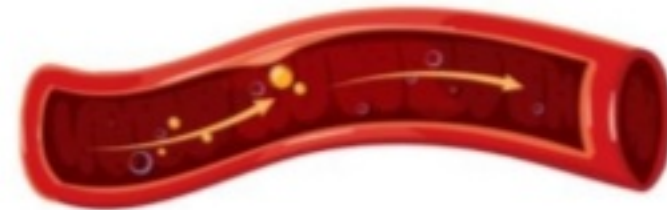


ATHEROSCLEROSIS:

A BUILDUP OF CHOLESTEROL PLAQUE IN THE WALLS OF ARTERIES, CAUSES OBSTRUCTION OF BLOOD FLOW.



NORMAL FUNCTIONS



ENDOTHELIAL DYSFUNCTION



PLAQUE FORMATION



PLAQUE RUPTURE THROMBOSIS

- 1) It is an abnormality associated with cholesterol metabolism. Blood cholesterol level is always high in atherosclerosis.
- 2) However, genetic factors are also involved in the development of this disease.
- 3) In this condition, initially cholesterol esters particularly cholesterol oleates of arterial smooth muscle cells deposits in arterial intima. This leads to fatty streaks formation and condition is reversible. If condition is not controlled continued extracellular deposition of cholesterol esters along with apo B-100 of lipoproteins results in the formation of plaque in the arterial wall.
- 4) Plaque formation in the arterial wall causes narrowing of arterial lumen.
- 5) Blood vessel narrowing due to deposition of cholesterol ester and apo B-100 is called as **atherosclerosis**.
- 6) Plaque in arteries promotes clot formation.



- 7) If clot formation occurs in coronary artery, the blood and O₂ supply to cardiac muscle diminishes. This manifests as myocardial infarction or stroke because anoxia causes necrosis of cardiac tissue.
- 8) Thus atherosclerosis cause coronary artery, disease (CAD)
- 9) Atherosclerosis may develop as secondary complication of diseases like diabetes, hypothyroidism, lipid nephrosis and other type of dyslipoproteinemias.
- 10) Some atherosclerotic lesions occurs even with normal blood cholesterol level. Inflammatory factors, low HDL levels are involved in this type of atherosclerosis development. Decreased HDL level leads to monocyte in filtration into arterial wall, macrophage, foam cell formation and lesion.

FATTY LIVER:

- Liver contains about 5% lipid. Of this, about 1/4 is triglyceride.
- Extensive accumulation of lipid in the liver leads to condition known as fatty liver. In the fatty livers, the lipid content increases to 25-30%. Further, triglycerides and fatty acid may occupy entire cytoplasm of hepatocyte.

SEVERAL FACTORS CAUSES ACCUMULATION OF LIPID IN LIVER:

1) RAISED PLASMA FREE FATTY ACID LEVEL:

When there is a mobilization of fat from adipose and extrahepatic tissues plasma free fatty acid level increases. Liver takes up increasing amounts of fatty acids and esterifies them. Hence rate of triglyceride synthesis is more. However, the synthesis of VLDL occurs only at normal rate. As a result triglycerides accumulate and cause fatty liver. The plasma free fatty acid level is elevated in (a) starvation (b) diabetes (c) high fat diet (d) carnitine deficiency. Hence, in these conditions fatty liver occurs.

2) METABOLIC BLOCK IN THE PRODUCTION OF LIPOPROTEINS:

- Block in VLDL formation causes fatty liver even though the rate of triglyceride synthesis is normal because VLDL transports triglyceride from liver to extra hepatic tissues. VLDL formation may be blocked if substance (s) required for its formation are deficient. However, when deficient substances are supplied fat accumulation cease.

- **Lipotropic factors** Are compounds that relieve or prevent excess accumulation of lipids in the liver. They are choline, methionine and betaine. These lipotropic factors cure fatty liver due to choline or methionine deficiency. Choline deficiency may result from impaired transmethylation reactions associated with methionine catabolism.
- Choline deficiency leads to block in choline dependent phospholipid biosynthesis. This in turn impairs formation of membranes needed for lipoprotein synthesis. Thus choline deficiency results in block in VLDL formation and causes accumulation of fat in liver. Choline deficiency may impairs carnitine biosynthesis also.
- Other noteworthy lipotropic factors are PUFA, vit E, pyridoxine and pantothenic acid. The deficiency of any one of these substances causes fatty liver. However, they can not prevent occurrence of fatty liver due to choline deficiency.

3) TOXIC SUBSTANCES:

Several hepato toxic agents like carbon tetrachloride, chloroform, phosphorus, lead, arsenic, alcohol and orotic acid causes fatty liver. Substances, which inhibit protein synthesis like puromycin and ethionine, a methionine analog also cause fatty liver.

MEDICAL IMPORTANCE:

EFFECT OF FATTY LIVER: When accumulation of lipid in liver becomes chronic fibrotic changes takes place in hepatocytes, which progress to cirrhosis and finally impaired liver function.

HYPERCHOLESTEROLMIEA:

- An elevated blood cholesterol level is called hypercholesterolemia. Throughout the body, cholesterol is found in all cells as a waxy substance. Typical sources of cholesterol are meat, poultry, dairy products, eggs and fish. Foods rich in saturated fats and trans fats cause the liver to make an excess of cholesterol, which can result in hypercholesterolemia in some cases.
- There are various functions of cholesterol in the human body, including the synthesis of cell membranes, some hormones, and substances needed for fat digestion. An excessive level of cholesterol, however, can lead to coronary artery disease.

Increase in plasma cholesterol (> 200 mg/dl) concentration is known as hypercholesterolemia and is observed in many disorders

- 1) **Diabetes mellitus:** Due to increased cholesterol synthesis since the availability of acetyl CoA is increased.
- 2) **Hypothyroidism (myxoedema):** This is believed to be due to decrease in the HDL receptors on hepatocytes.
- 3) **Obstructive jaundice:** Due to an obstruction in the excretion of cholesterol through bile.
- 4) **Nephrotic syndrome:** Increase in plasma globulin concentration is the characteristic feature of nephrotic syndrome. Cholesterol elevation is due to increase in plasma lipoprotein fractions in this disorder.

Hypercholesterolemia is associated with **atherosclerosis** and **coronary heart disease** (CHD). More specifically, LDL-cholesterol is positively correlated, whereas HDL-cholesterol is negatively correlated with CHD.

BAD CHOLESTEROL AND GOOD CHOLESTEROL:

Cholesterol is a natural metabolite performing a wide range of functions (membrane structure, precursor for steroid hormones, bile acids). The usages good and bad to cholesterol, although inappropriate, are still in use. The cholesterol in high concentration, present in LDL, is considered bad due to its involvement in atherosclerosis and related complications. Thus, LDL may be regarded as lethally dangerous lipoprotein. Small dense LDL (sdLDL) is considered to be the most dangerous fraction of LDL associated with CHD. On the other hand, HDL cholesterol is good since its high concentration counteracts atherogenesis. HDL may be considered as highly desirable lipoprotein.

AFFECTS OF LIFESTYLES ON SERUM CHOLESTEROL LEVEL:

Individual lifestyles and habits certainly influence serum cholesterol, and thus play a significant role in the development coronary heart disease. The parameters such as high blood pressure, emotional stress, smoking, drinking of soft water (against hard water), coffee drinking, lack of exercise, obesity (particularly of abdomen) elevate serum cholesterol level.